Global Noise Reduction versus Phonetic Preservation in Wiener Speech Enhancement

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Project site: https://saotao117.github.io/elec5305-project-550603890/

Code: https://github.com/SAOTAO117/elec5305-project-550603890

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Abstract

This study examines phonetic preservation in Wiener speech enhancement. A short-time Fourier transform (STFT) filter was tested on one labelled English utterance from each of ten speakers. Each utterance was mixed with three environmental noises at four signal-to-noise ratios (SNRs). Two noise excerpts per utterance–noise pair gave 240 mixtures. The baseline used a 32 ms window. Applying the same gain to the clean speech and noise separated speech attenuation from noise suppression. Mean error SNR improved by 6.18 dB, but mean short-time objective intelligibility (STOI) fell from 0.8432 to 0.8385. With equal speaker weights, speech retention averaged −1.24 dB for vowels, −2.53 dB for fricatives and −2.51 dB for stops and affricates. Other measures gave different class rankings. A paired 16 ms experiment reduced frames crossing phone boundaries but increased speech attenuation and lowered STOI. These results show unequal signal preservation in this sample. Whether these changes affect the perception of specific speech sounds remains untested.

1 Introduction

A Wiener filter attenuates spectral components whose estimated speech power is low relative to noise power. A weak consonant may therefore be attenuated along with the noise. Vowels, fricatives and stops differ in energy, periodicity and duration. A whole-sentence score can hide these differences because strong vowels contribute more to total speech energy than short, weak consonants.

The research question is whether improved global error SNR can hide unequal preservation of phonetic classes. Controlled mixtures and phone timing labels allow the same filter to be evaluated for each class. Here, a phone is a labelled occurrence of a speech sound. The analysis compares gain, speech and noise retention, spectral distortion and segmental error SNR. It tests which classes are most affected without assuming that fricatives suffer the greatest loss.

Studies of gain distortion [2] and multichannel enhancement [3] support evaluating speech sounds separately. They do not directly answer this question for the selected single-channel Wiener filter. This project applies a standard filter and compares its effects at utterance and phone levels. A paired window-length experiment then tests whether shorter analysis frames improve speech preservation.

2 Literature Review

2.1 Enhancement and speech distortion

Loizou [1] derives the Wiener estimator by minimising mean-square error. Its gain is the ratio of speech power to total spectral power. This provides an interpretable baseline, but its performance depends on the power estimates. Weak speech and changing noise can make the local SNR estimate inaccurate. A gain floor limits attenuation without accounting for the importance of a speech sound. The present study uses one fixed configuration so that its effects on phonetic classes can be examined directly.

Kim and Loizou [2] separated amplification and attenuation distortions in listening experiments. Attenuation caused little intelligibility loss under their test conditions. Signals containing only attenuation distortion could be much more intelligible than the unprocessed noisy speech. Energy loss alone therefore cannot establish a loss of speech information. This finding supports measuring speech attenuation and residual noise separately, alongside an intelligibility measure. The present study uses component energy and STOI. It does not reproduce their listening experiment or square-root Wiener rule.

2.2 Phonetic structure and evaluation

Monir, Magron and Serizel [3] evaluated three multichannel systems at phoneme level. Their results showed differences between categories, including difficulty with plosives and benefits for some nasals and sibilants. These findings support local evaluation but do not establish a universal ranking of vulnerable sounds. Their systems use spatial information, and their room conditions and metrics differ from those used here. The effect of the selected single-channel Wiener gain therefore needs to be tested under the present noise conditions.

Kong, Mullangi and Kokkinakis [4] classified fricatives using short-frame spectral features and tested their robustness to noise. Their results show the value of spectral shape above 1 kHz for distinguishing fricatives. Shadle and colleagues [5] also show that fricative measurements depend on the analysis method and the frequency range. Together, these studies support measuring frequency bands separately. The present recordings have a 16 kHz sampling rate, which limits analysis to 8 kHz. The high-frequency measure therefore covers 4–8 kHz and omits any fricative information above that range.

STOI uses short-time spectral envelopes to predict intelligibility [6]. It provides an utterance-level reference but cannot identify which phone caused a score change. Error SNR, spectral distortion and speech retention measure other aspects of performance. This study uses them together to assess signal preservation. None provides a direct count of correctly perceived phonemes.

2.3 Method selection

Spectral subtraction is a simple alternative, although it can leave musical-noise artefacts [1]. Multichannel methods use spatial information but require several microphone signals [3]. The Wiener power-ratio gain in [1] suits the available single-channel recordings and can be inspected directly. It was selected for this reason. Its main limitation remains the accuracy of the power estimates, especially for weak speech and changing noise.

Controlled mixing gives the exact clean speech and noise needed for component analysis. The speech comes from LibriSpeech [7], with phone boundaries supplied by the derivative dataset [9]. DEMAND provides recorded environmental noise [8]. Using existing boundaries avoids developing a new alignment system, although the supplied labels have not been manually reannotated. MATLAB supplies STFT and inverse-STFT operations [10]. The project code implements mixture generation, gain estimation and phonetic measurements.

Component analysis requires the same gain for speech and noise. Re-estimating the gain on each clean reference would change the filter and invalidate the decomposition. Frame assignment requires a separate decision. Assigning each frame by its centre avoids double counting, but the window may still include an adjacent phone. A numerical check verifies the decomposition, and the window-length experiment examines this boundary effect.

3 Methods

3.1 Data and experimental conditions

The sample uses the first eligible utterance from each of ten speakers in the dev.clean split of [9]. Each recording is 16 kHz mono and lasts 3–8 s. It must contain all four speech classes and silence, with known phone symbols and continuous labels. The 549 segments include 192 vowels, 113 nasals and approximants, 104 fricatives, 97 stops and affricates, and 43 silence segments. Fricatives include voiced sounds and HH. Stop labels cover the full segment rather

than an isolated burst. Silence provides a separate gain control and is excluded from speech-retention comparisons.

The noise recordings are kitchen noise (DKITCHEN), cafeteria noise (PCAFETER) and traffic noise (STRAFFIC). Only the first channel is used. A fixed random seed selects two excerpts for each utterance–noise pair. Each excerpt is reused at −5, 0, 5 and 10 dB, giving 240 mixtures per configuration. Excerpts may overlap. They provide repeated noise conditions, not additional independent speakers.

Table 1. Filter settings and controlled window comparison.

Parameter	Baseline	Comparison
Sampling rate	16 kHz	16 kHz
Window	512 samples, 32 ms	256 samples, 16 ms
Analysis window	Square-root periodic Hann	Same window family
Hop and overlap	256 samples, 50%	128 samples, 50%
FFT length	512 points	512 points
Mixture-power smoothing	0.8	0.8944
Noise estimate	0.5 s noise-only prefix	Same prefix
Minimum amplitude gain	0.05	0.05

The prefix estimate, power smoothing and gain floor are implementation choices for the Wiener rule. A smoothing factor of 0.8 reduces power fluctuations and gives a time constant of about 72 ms. The 0.05 gain floor prevents spectral nulls and limits per-bin attenuation to about 26 dB. These settings define the baseline but have not been optimised for this dataset.

3.2 Mixture generation and Wiener filtering

Let s[n] be the clean speech and v[n] the selected noise. A denotes the samples labelled as active speech. Sums indexed by A or I include only samples in that set. The mixture and noise scale are

$$y[n] = s[n] + \alpha v[n], \quad \alpha = \left[\frac{\sum_A s[n]^2}{\sum_A v[n]^2 \cdot 10^{\mathrm{SNR}/10}}\right]^{1/2} \tag{1}$$

Each utterance has a 0.5 s noise-only prefix. Input SNR calibration uses the active samples in A. Whole-utterance evaluation excludes the prefix but includes labelled silence. Its input error SNR can therefore differ from the target SNR. The scaled noise is n[n] = αv[n].

Let x be the signal after boundary zero padding and w the analysis window. L is the window length, H the hop and K the transform length. The indices k, m and l denote the frequency bin, frame and sample within the window, respectively. With imaginary unit j, the STFT is

$$X[k,m] = \sum_{l=0}^{L-1} x[mH+l]w[l] \exp(-j2\pi kl/K) \tag{2}$$

Y, S and N are the STFTs of the mixture, clean speech and scaled noise. Padding and prefix offsets are removed from frame-centre times to match the utterance labels. The noise estimate $P_n[k]$ averages $|Y[k,m]|^2$ over windows wholly inside the prefix. This estimate remains fixed during speech. Mixture power is initialised from $P_n[k]$ and smoothed with β = 0.8:

$$P_y[k,m] = \beta P_y[k,m\text{-}1] + (1\text{-}\beta)|Y[k,m]|^2 \tag{3}$$

$$P_s[k,m] = \max(P_y[k,m] - P_n[k], 0) \tag{4}$$

$$G[k,m] = \max(0.05,\ \frac{P_s[k,m]}{\max(P_s[k,m] + P_n[k],\ \varepsilon)}) \tag{5}$$

The numerical floor ε is the larger of 10^{-12} times the maximum mixture STFT power and realmin. For uncorrelated speech and noise, minimising $E\{|S-GY|^2\}$ gives the ideal gain $P_s/(P_s+P_n)$ [1]. Equations (3)–(5) use estimated powers and a lower gain bound to approximate this rule. Speech power is estimated by subtracting the fixed noise power from the smoothed mixture power. Gain estimation uses only the noisy input and its prefix. Clean speech is used only for evaluation.

The enhanced spectrum GY retains the mixture phase. Inverse STFT (ISTFT) uses weighted overlap-add with the same window, normalised by the sum of squared window values. The square-root Hann window and 50% overlap support unity-gain reconstruction. After removing boundary padding, the enhanced signal ŷ and its components are

$$s_G = \mathrm{ISTFT}(GS), \quad n_G = \mathrm{ISTFT}(GN), \quad \hat{y} = s_G + n_G \tag{6}$$

3.3 Phonetic measurements

Each STFT frame is assigned once to the phone containing its centre. For a phone with exact sample interval I, the primary speech and noise retention measurements are

$$R_s(I) = 10 \log_{10} \frac{\sum_I s_G[n]^2}{\sum_I s[n]^2}, \quad R_n(I) = 10 \log_{10} \frac{\sum_I n_G[n]^2}{\sum_I n[n]^2} \tag{7}$$

More negative values indicate stronger attenuation. Both ratios use the same sample interval after synthesis. Overlap-add can move energy across a short interval, so occasional positive speech-retention values are retained. A separate frame-energy ratio is used to check boundary effects. The 4–8 kHz measure uses clean STFT energy and the same gain in the assigned frames. One-sided energy weights are two, except at DC and Nyquist.

Unweighted gain averages G across one-sided bins and assigned frames. Energy-weighted gain uses $|S|^2$ and the one-sided factors as weights. The latter measures amplitude gain, while frame-energy retention depends on squared gain. Both differ from retention measured over exact sample intervals after synthesis. Gain is also averaged below 1 kHz, from 1–4 kHz and from 4–8 kHz.

$$\mathrm{SNR}_e(I,z) = 10 \log_{10} \frac{\sum_I s[n]^2}{\sum_I (z[n]-s[n])^2} \tag{8}$$

The error-SNR change is $\mathrm{SNR}_e(I,\hat{y}) - \mathrm{SNR}_e(I,y)$. The full utterance gives the global score, and individual phone intervals give segmental scores. Output error includes both speech modification and residual noise. Log-spectral distortion (LSD) is calculated as

$$\mathrm{LSD} = \frac{1}{M} \sum_{m=1}^{M} \left[\frac{1}{B} \sum_{k=0}^{B-1} (D_{\text{processed}}[k,m] - D_{\text{clean}}[k,m])^2 \right]^{1/2} \tag{9}$$

M counts assigned frames and B counts one-sided bins. D is log magnitude, calculated with 20 $\log_{10}$ after flooring magnitudes at 10^{-3} times the maximum clean STFT magnitude. Mixture LSD compares GY with S, and speech-component LSD compares GS with S. STOI uses the full utterance. Speech metrics exclude segments below 10^{-12} of utterance speech energy. High-frequency ratios require band energy above 10^{-10} of the total weighted clean energy in the assigned STFT frames. Spectral metrics without frames are recorded as NaN, and valid counts are retained.

3.4 Aggregation and validation

Repeat measurements are averaged within each phone, then across phones for each speaker, noise, SNR and class. Class summaries weight speakers equally. Pooled results first average

conditions within speaker. Energy ratios are averaged in dB, not calculated from pooled samples. Error bars show between-speaker standard deviations. Paired contrasts subtract each speaker's vowel score in the same condition. Percentile intervals use 10,000 resamples of the ten paired differences, with seed 5305. These exploratory intervals are not adjusted for multiple comparisons.

The maximum relative reconstruction and decomposition errors are 2.70×10^{-16} and 3.03×10^{-16}. Further checks test unity gain, gain limits and independence from clean reference components. They also check one-sided energy weights and missing-frame handling. Active SNR is verified for every mixture. All 549 tokens receive at least one baseline frame. However, 40 of the 506 speech tokens have no 32 ms frame wholly inside the label.

The paired 16 ms experiment uses the same speech, labels, noise crops and scaling. Halving the hop and setting β to the square root of 0.8 preserves the smoothing time constant. A 512-point FFT keeps the same frequency grid through zero padding. True frequency resolution still changes with window length. The comparison measures time-domain retention, global error SNR, STOI and the fraction of frames crossing phone boundaries.

4 Results

4.1 Whole utterance performance

Mean global error SNR improves in all twelve noise–SNR conditions (Table 2). The pooled improvement is 6.18 dB. Across the four SNRs, improvements average 7.29 dB for kitchen noise, 4.39 dB for cafeteria noise and 6.86 dB for traffic noise. These differences apply to the selected recordings. Noise stationarity was not measured, so it cannot be identified as the cause.

Table 2. Global error-SNR improvement and STOI change. SNR values are mean ± SD across ten speakers after averaging the two noise excerpts.

Noise	Input SNR (dB)	Error-SNR change (dB)	STOI change
Kitchen	-5	7.54 ± 1.94	+0.0060
Kitchen	0	7.42 ± 1.89	+0.0024
Kitchen	5	7.24 ± 1.83	+0.0007
Kitchen	10	6.97 ± 1.78	+0.0001
Cafeteria	-5	5.29 ± 0.58	-0.0389
Cafeteria	0	4.69 ± 0.58	-0.0191
Cafeteria	5	4.09 ± 0.63	-0.0068
Cafeteria	10	3.51 ± 0.68	-0.0003
Traffic	-5	7.88 ± 2.28	-0.0002
Traffic	0	7.18 ± 1.94	-0.0001
Traffic	5	6.49 ± 1.75	-0.0004
Traffic	10	5.90 ± 1.73	+0.0004

Mean STOI decreases from 0.8432 to 0.8385. Cafeteria noise gives the largest decrease, −0.0389 at −5 dB, despite an error-SNR improvement of 5.29 dB. In this condition, reducing the signal error does not improve the objective intelligibility score.

4.2 Speech retention and class differences

Table 3. Pooled phone measurements. Energy ratios are mean ± SD across ten speaker means, with equal weight for noise and SNR conditions.

Class	Speech retention (dB)	Noise retention (dB)	Mean gain	Weighted gain
Vowels	-1.24 ± 0.31	-8.67 ± 1.88	0.450	0.875
Nasals and approximants	-1.48 ± 0.51	-8.84 ± 2.13	0.424	0.851
Fricatives	-2.53 ± 1.14	-9.02 ± 1.90	0.465	0.749
Stops and affricates	-2.51 ± 0.85	-9.01 ± 2.11	0.422	0.742

Fricatives and stops or affricates lose more speech energy than vowels in the pooled results (Table 3). Their noise retention is only slightly lower. All four classes undergo substantial noise suppression, but the amount of desired speech removed differs. Figure 1 shows this contrast across input SNRs in cafeteria noise. Retention approaches zero as the mixture becomes cleaner.

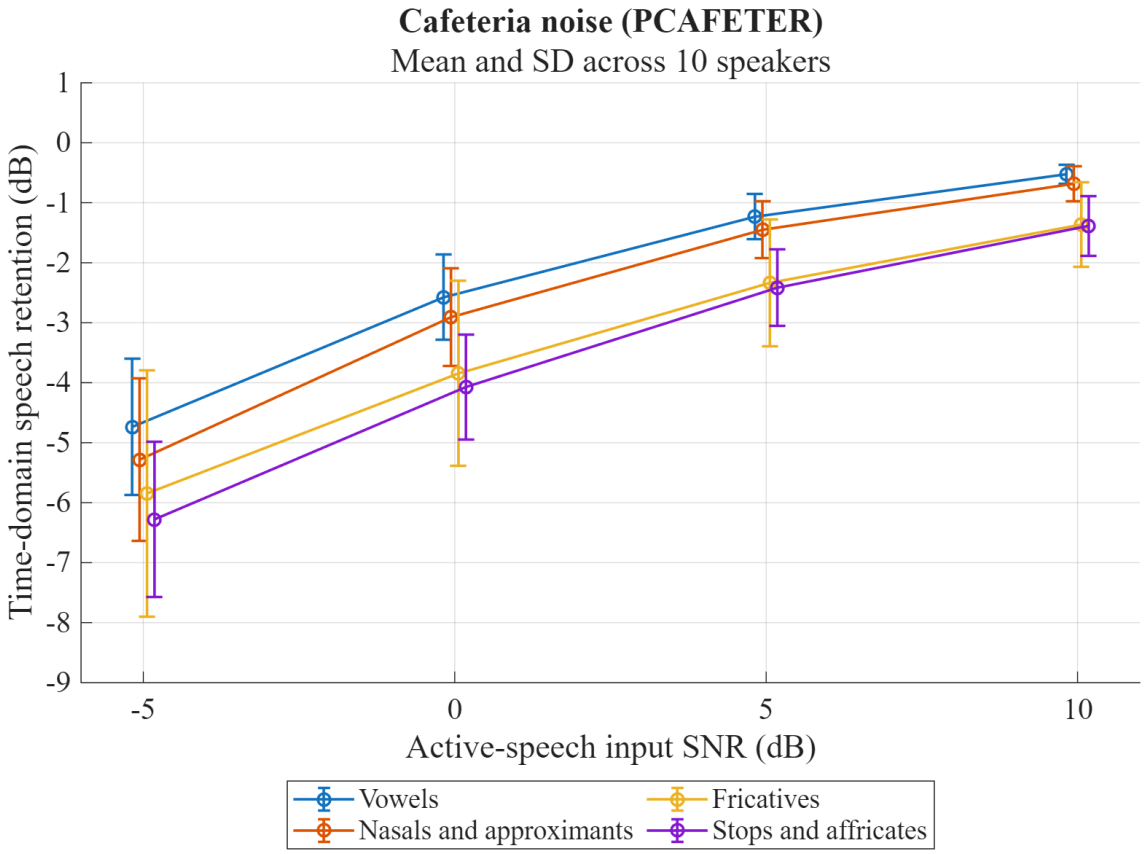


Figure 1. Time-domain speech retention in cafeteria noise. Bars show SD across ten speaker means. Horizontal offsets separate the class markers; all classes use the same four input SNRs.

At −5 dB in cafeteria noise, speech retention is −4.74 dB for vowels, −5.85 dB for fricatives and −6.28 dB for stops and affricates. The paired fricative–vowel difference is −1.11 dB, with a 95% interval of [−2.08, −0.25]. The corresponding stop/affricate–vowel difference is −1.54 dB, with an interval of [−2.29, −0.80].

Mean segmental error SNR improves for every combination of speech class, noise and SNR. At −5 dB, the noise-averaged improvement is 8.17 dB for vowels and 8.66 dB for nasals and approximants. It is 9.50 dB for fricatives and 9.57 dB for stops and affricates. The larger improvements in the latter groups occur alongside greater speech attenuation.

4.3 Gain and spectral preservation

Across all conditions, unweighted mean gain is 0.465 for fricatives and 0.450 for vowels (Table 3). Energy weighting reverses this ranking: the values are 0.749 and 0.875, respectively. The unweighted mean includes bins with little clean speech energy. At 0 dB input SNR, noise conditions are averaged within each speaker. Mean gain is then 0.248 for silence, below 0.395 for vowels and 0.411 for fricatives. This confirms stronger suppression during non-speech intervals.

Below 1 kHz, at 1–4 kHz and at 4–8 kHz, pooled fricative gains are 0.536, 0.446 and 0.461. Vowel gains are 0.615, 0.496 and 0.374 in the same bands. The ranking reverses in the highest band. A single average gain therefore hides differences across frequency.

Pooled mixture LSD is 6.45 dB for vowels and 6.82 dB for fricatives. Speech-component LSD is 4.43 and 3.74 dB, respectively, so the class ranking depends on the metric. Frame-based retention in the 4–8 kHz band is −6.08 dB for fricatives and −6.91 dB for vowels. It is −8.06 dB for nasals and approximants and −7.55 dB for stops and affricates. This comparison concerns classes within one band. It does not compare high- and low-frequency losses within fricatives.

Figure 2 shows different gains across frequency within the same phonetic interval. Gain is mostly low during the final silence after about 6 s. During speech, some bands have gain close to one. Phonetic class alone therefore does not determine the gain applied to every bin.

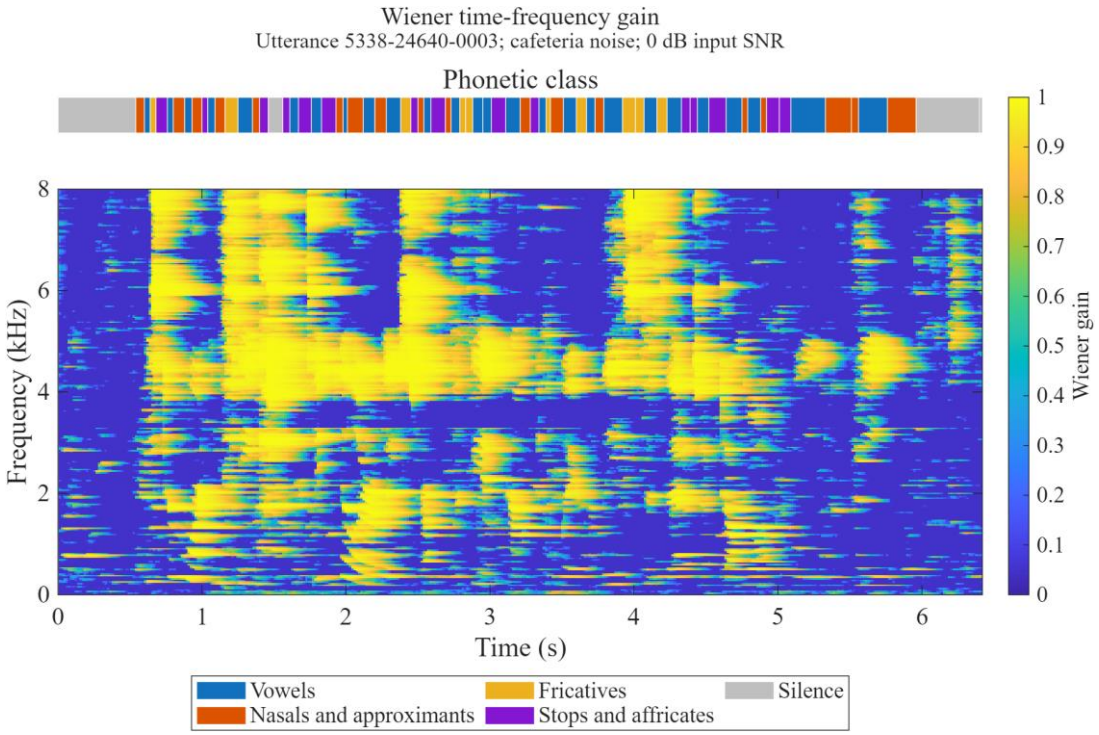


Figure 2. Wiener gain for utterance 5338-24640-0003 in cafeteria noise at 0 dB. The phonetic-class strip shares the time axis of the 0–8 kHz gain map.

In Figure 3, the enhanced vowel retains the main peaks below 1 kHz and around 4.5–5 kHz. Several weaker components between about 2.5 and 4 kHz fall below the clean spectrum, showing speech attenuation. The enhanced /s/ spectrum in Figure 4 follows the clean curve above 4 kHz. Across much of 1–3.5 kHz, it remains above the clean reference because of residual interference. This example agrees with the relatively good 4–8 kHz fricative retention. It also shows that preserving one band does not ensure low distortion across the full spectrum. These longer examples do not represent the shortest consonants.

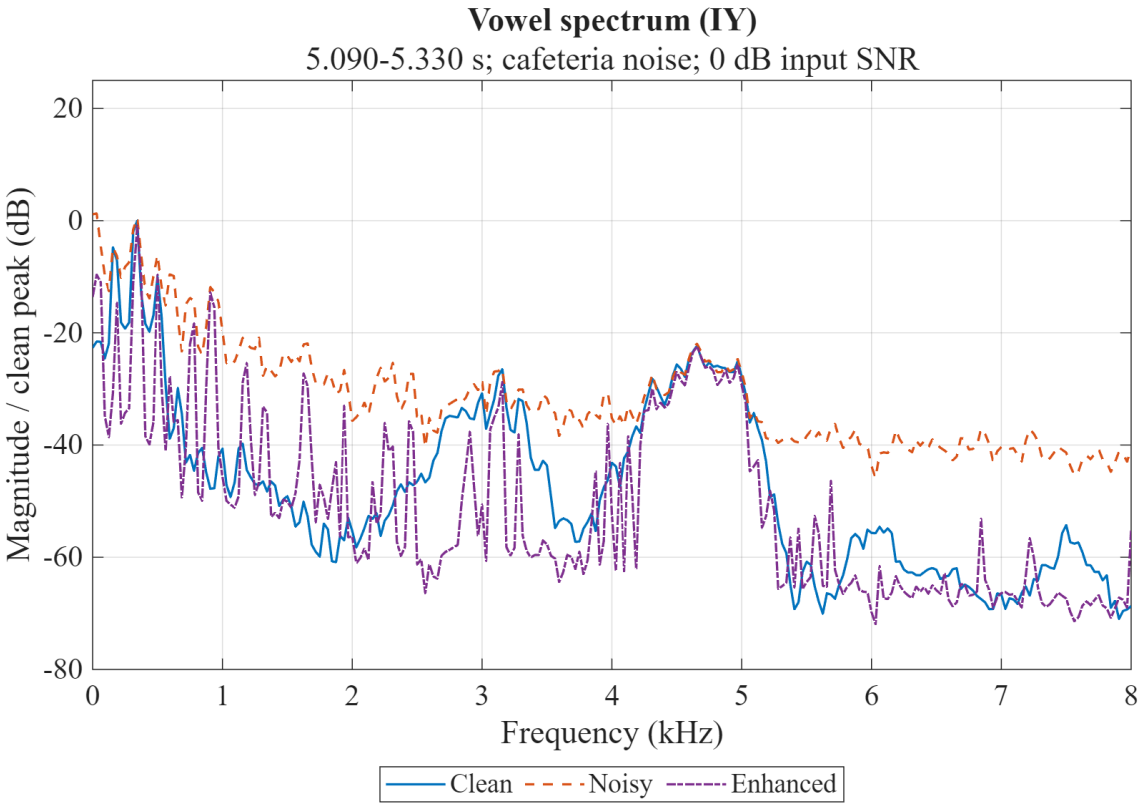


Figure 3. Clean, noisy and enhanced spectra for IY at 5.09–5.33 s. For Figures 3 and 4, the phone with the most assigned frames in each class is selected. RMS magnitudes share the clean spectral maximum as their reference.

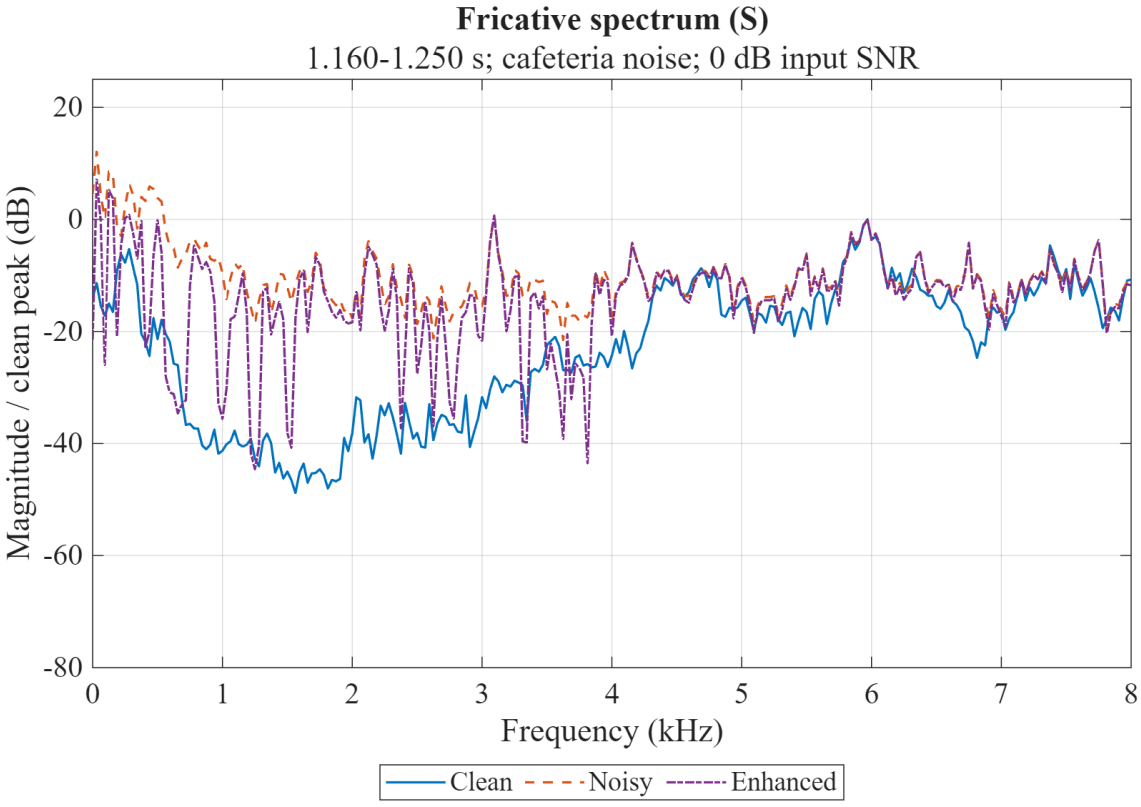


Figure 4. Corresponding spectra for the fricative labelled S at 1.16–1.25 s. High-frequency structure is comparatively well retained in this example, while the enhanced spectrum still differs from the clean reference at lower frequencies.

4.4 Window length comparison

Table 4. Paired changes from 32 ms to 16 ms, pooled within speaker. Intervals are exploratory 95% speaker-bootstrap intervals.

Measurement	Mean change	95% interval
Global error-SNR improvement (dB)	+0.063	[-0.043, +0.166]
STOI	-0.0093	[-0.0110, -0.0077]
Global noise retention (dB)	-0.362	[-0.491, -0.249]
Vowel speech retention (dB)	-0.477	[-0.561, -0.396]
Nasal/approximant speech retention (dB)	-0.609	[-0.768, -0.474]
Fricative speech retention (dB)	-0.593	[-0.711, -0.486]
Stop/affricate speech retention (dB)	-0.628	[-0.767, -0.484]

Shortening the window reduces the class-average fraction of boundary-crossing frames from about 42–46% to 19–21%. However, speech retention falls in every class, and STOI also decreases (Table 4). Global error-SNR improvement changes by only +0.063 dB, with an interval that includes zero. The shorter window suppresses more noise but also removes more speech. The 32 ms configuration is therefore retained as the baseline. Its performance on other data or settings remains untested.

5 Discussion

5.1 Interpretation of the main findings

The tested Wiener filter preserves different phonetic classes unequally in energy terms. The clearest contrast is between vowels and the fricative or stop/affricate groups under difficult noise conditions. Global and segmental error SNR can improve at the same time. Thus, an improved error score can hide greater attenuation of particular speech classes. This answers the signal-level part of the research question. The effect on phoneme or word perception remains untested.

Enhanced-speech error contains both speech modification and residual noise. Its squared norm includes their energies and a cross-term. Removing enough noise can therefore reduce the error even when more speech is attenuated. Global scores also give greater weight to strong speech regions, while class scores average individual phone measurements. These different weightings help explain why the measures give different values and class rankings.

Energy-weighted gain helps explain the limitations of an unweighted mean. Fricatives can have high gain in many weak bins while losing energy in stronger bins. Yet their pooled 4–8 kHz retention is less negative than that of the other classes. This ranking applies only to that band. It does not describe attenuation across frequencies within a fricative. The individual spectra illustrate local effects, which may differ between tokens.

5.2 Relation to previous work

Monir and colleagues [3] also found differences between utterance-level and phoneme-level outcomes. Their reported benefits for some sibilants agree with the finding here that fricatives are not disadvantaged on every measure. Direct numerical comparison is not possible because the systems, conditions and metrics differ. The present study adds a controlled single-channel example. Its component analysis uses the same gain for speech and noise to separate their contributions to the output.

Kim and Loizou [2] show why attenuation should not automatically be interpreted as an intelligibility loss. Here, speech energy and mean STOI both decrease, but this does not establish a causal link. Kong and colleagues [4] and Shadle and colleagues [5] support examining spectral structure. A broad band-energy ratio captures only part of that structure and cannot assess every cue to fricative identity. The 8 kHz upper limit also excludes information at higher frequencies.

5.3 Window length and signal preservation

Shorter windows include less of the adjacent phone, but the experiment also shows greater speech attenuation and lower STOI. Window length changes more than boundary mixing. It also changes true frequency resolution and the sequence of power observations, even with the same smoothing time constant. These effects cannot be separated by this comparison. The small improvement in error SNR does not justify replacing the 32 ms baseline when speech preservation is the priority.

5.4 Limitations and further work

The study uses a small convenience sample of ten speakers, with one utterance per speaker. Equal speaker weights prevent differences in token counts from dominating the results. They do not control for duration, speaking style, original energy or phonetic context. The supplied alignments were checked for structural errors but not manually verified. Broad classes combine different sounds, and stop intervals include more than the burst. Bootstrap intervals remain exploratory, particularly when many conditions are compared.

Each noise environment is represented by one recording. The noise estimate is fixed, so later changes in the noise spectrum or level may affect performance. The experiment does not measure stationarity and cannot attribute the cafeteria result to one noise property. The gain floor and power smoothing were not exhaustively tuned. LSD depends on the magnitude floor, and energy ratios can be unstable for nearly silent segments. Thresholds reduce this problem, but the results still depend on how the metrics are defined.

Further work should include several utterances per speaker and manual checks of short consonant boundaries across classes and durations. Noise estimation or gain-floor settings could then be compared using a separate development set. Evaluation on held-out recordings should retain the paired speech and noise measurements. Listening tests would be needed to determine whether the observed changes affect phoneme or word recognition.

6 Conclusion

Across 240 controlled mixtures, the baseline improved mean whole-utterance error SNR by 6.18 dB, while mean STOI fell slightly. Applying the same gain to speech and noise revealed greater pooled speech attenuation for fricatives and stops or affricates than for vowels. Unweighted gain and high-frequency retention gave different rankings. The metrics therefore need to be considered together when assessing phonetic preservation. The 16 ms window reduced boundary mixing but worsened speech retention and STOI. The 32 ms configuration remains the baseline. These findings establish unequal signal preservation in the selected data. Perceptual effects and generalization to a wider population require further testing.

Reproducibility

The code was tested with MATLAB R2025b and Signal Processing Toolbox. Audio Toolbox provides STOI. The project includes the selected audio, labels and manifests. Running main rebuilds the baseline. The function validateDesign checks the signal path and metrics, while compareWindows runs both configurations and their paired comparison. Project-site and repository links appear on the first page. Result tables retain individual conditions, speaker means and calculation settings.

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